Axam: AI-Powered Education Without Boundaries

When There Is No Internet: AI-powered EdTech platform with offline capabilities.

Why?: Its personal.

axamai.org

E by Emmanuel Kasigazi

Purpose & Creation

Tech Divide in Education: Who's Left Behind in the AI Age?

	Underserved Regions	**Developed Cities**
Internet Access	Less than 30% of schools connected	More than 95% of schools connected
Device Access	1 in 5 students	1:1 device-to-student ratio
AI Integration	Less than 5% exposure in classrooms	30–60% use AI-enhanced tools
Teacher Readiness	Less than 10% trained on EdTech	Ongoing digital PD & AI training

1.3B+ children lack internet at home (UNESCO, 2023). 50% of low-income families and 42% of families of color lack technology required for online education.

AXAM

Description

AXAM is an offline First AI-powered learning platform. Developed through the NSF Innovation Corps (I-Corps) program, AXAM packages high-quality open educational resources, beginning with MIT OpenCourseWare transcripts and expanding to broader OER collections, into a locally deployable system that runs entirely without an internet connection.

1 Pre-trained AI Models

Pre-trained AI models optimized for low-resource environments.

Content works without internet, updates periodically when connected.

2 Hybrid

Hybrid delivery model combining software platform and hardware component.

Online development, offline deployment through portable devices

Multi-Lingo

3 Universal Access

- Partnership-driven distribution network
Deploy through educational institutions, NGOs, and government programs

- Scalable manufacturing and content pipeline
Mass-produce devices while continuously improving AI-generated materials

Founder Expertise

Emmanuel Olimi Kasigazi

- **15+ years building tech solutions in African markets:** Deep understanding of connectivity challenges and educational barriers.
- **Proven track record scaling businesses in emerging markets:** Led Wazi Group to 80% revenue growth, managed 1.5M+ contracts across East Africa.
- **MIT partnership validates global education expertise:** Co-host Massachusetts Institute of Technology OpenCourseWare podcast showcasing learners use of OER worldwide.
- **Technical background in AI/ML:** Currently pursuing an M.S. in Data Analytics Yeshiva University's Katz School, where I also serves as President of the KASA, representing over 800 STEM graduate students.
- **National Science Foundation I-corps** fellowship (New York I-Corps Hub Regional with the NYC Innovation Hot Spot)
- Linkedin.com/olimiemma:

Target Market

Educational Institutions

Schools, universities,and

training centers

Governments & NGOs

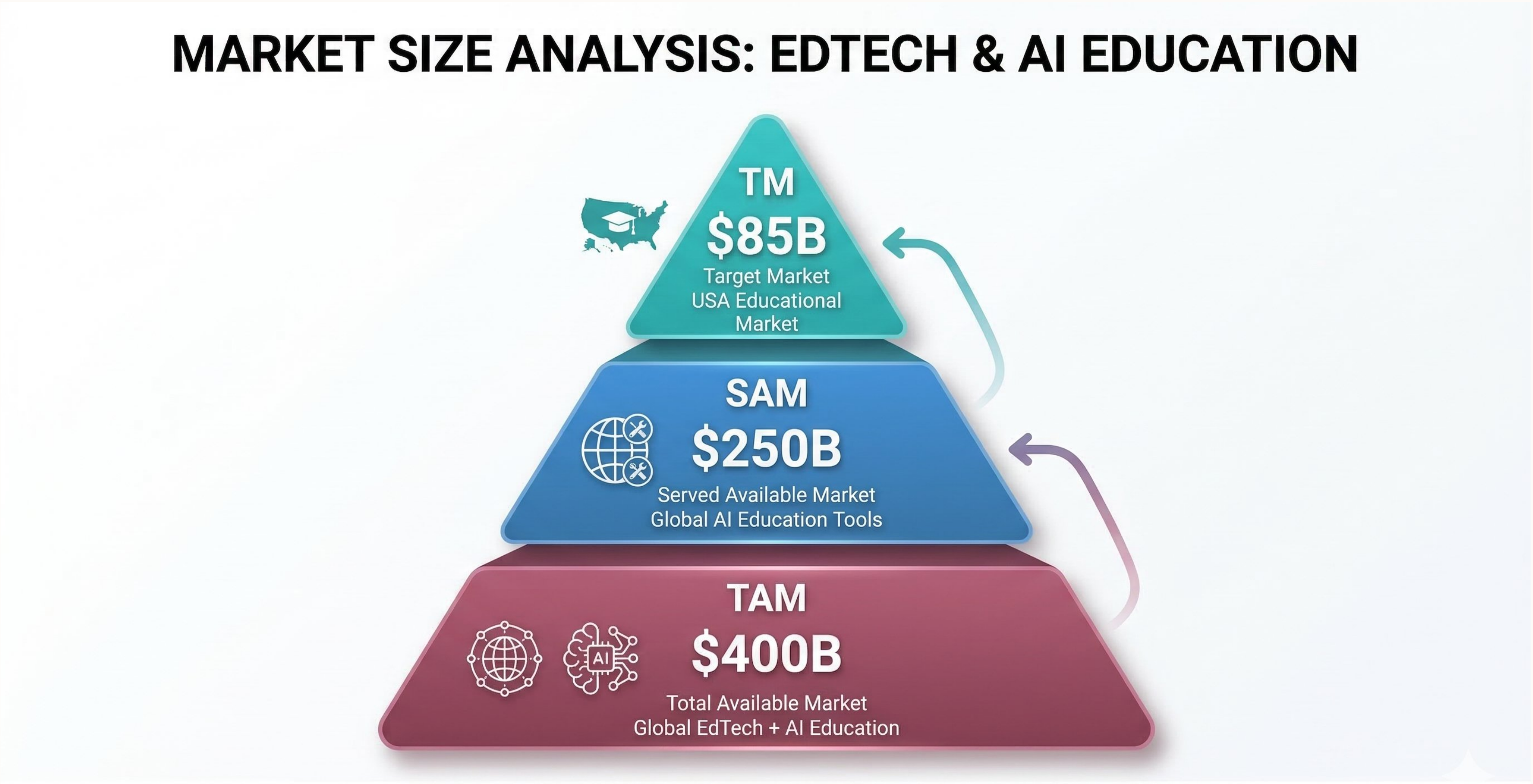
Government Education

Departments and Development

Organizations

Direct Customers

Parents in their homes,
Life Long learners



Competitive Landscape

StudyFetch	AI learning platform lacking offline capabilities
Educato AI	Focuses on underserved exams but requires internet
Mindgrasp	Offers AI-powered study tools but online-only
PSI Exams	Limited AI test prep for specific certifications
Traditional Materials	Static content without adaptive capabilities

HOW AXAM DIFFERS

- **Universal Coverage:** Targets underserved markets others ignore
- **Only AI-powered platform that works offline as well**
- **Universal coverage vs. niche competitors**
- **Adaptive learning with detailed explanations tailored to user needs.**

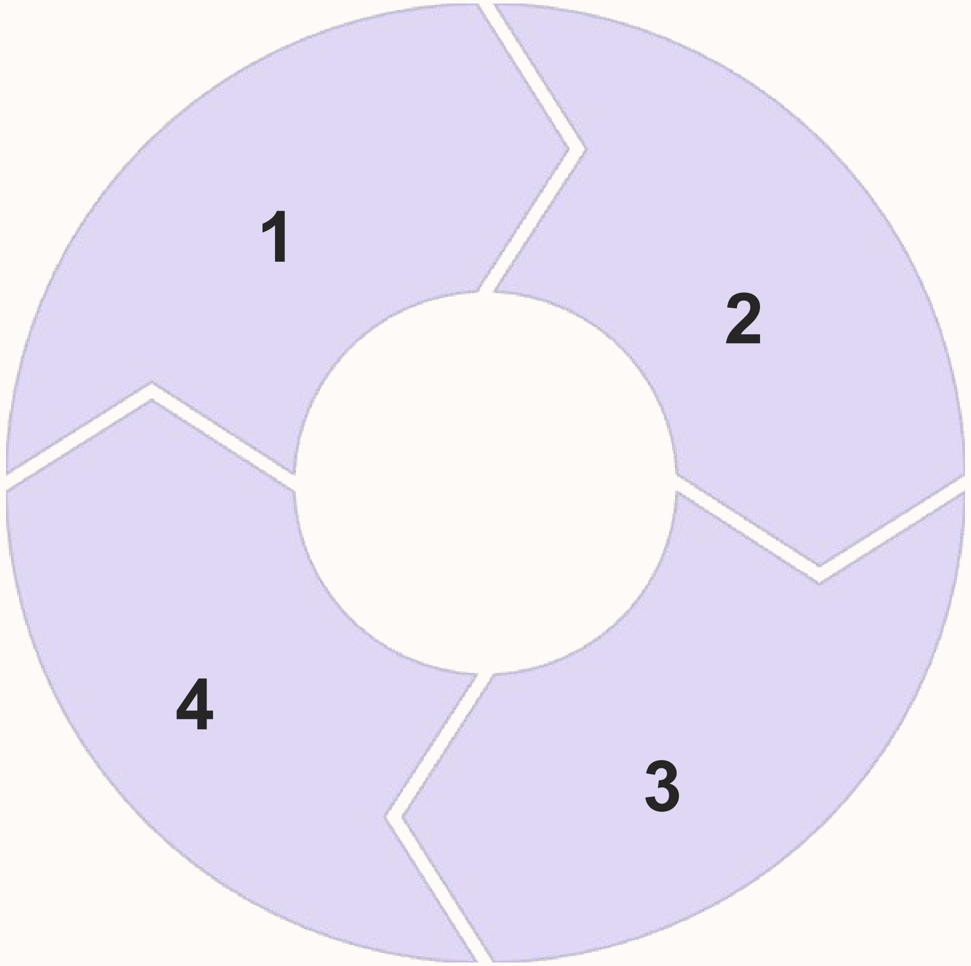
Business Model

Software Platform

Online component with subscription/freemium model

Hardware + Software Revenue Streams

Device sales/leasing + subscription model creates multiple income sources



Update System

Once in a while connectivity required for content refresh

Institutional Partnerships

Bulk licensing to schools, NGOs, and government education departments e.g schools or training centers

Getting Out of the Building

Who We Talked To (Interviews):

- NYC DOE Principals & Administrators (4)
- Teachers & Educators (4)
- EdTech Entrepreneurs & Workforce Development (3)
- IT Directors & Security Officers (4)
- Students & Parents (4)

Key Organizations: NYC DOE • District 79 Alternative Schools • Peblink • UVA • Success Academy • OEG

"We started with a technology looking for a problem. Customer discovery taught us to find the problem first."

Strategic interviews with real players in the education ecosystem:

NYC Department of Education
- Dr. Timothy Lisante (Retired Executive Superintendent, ACCESS Schools - 70,000 students)
- Joan Mosely (Principal, NYC public school - 92% economic need)

University Stakeholders
- Yeshiva University IT Leadership Team (CIO, Security Manager)
- UVA Research (Kaylyn Feeley, Autism Communication Research)

Industry Experts
- Sally Ricker (Founder, Peblink - Project-based learning & recruiting)
- Shiva Bhaggan (CEO, DivyAI - GovTech AI solutions)
- Alan Levine (Director of Innovation, Open Education Global)

Business Model Canvas

Key Partners

NYC Department of Education: As a regulatory and procurement partner.

EDUCAUSE is identified as an influential organization that sets standards for security vetting. They are a new, key player in

Colibri and Open Education Global to accelerate our entry into the mkt. We must partner with existing offline education pltfm

'local champions' (teachers, community leaders) in each region for successful implementation. We must establish partnerships

THE PRINCIPAL IS THE ECONOMIC BUYER WITH AUTONOMY.

Key Activities

Business: Navigating the NYC DOE vendor approval and ERMA procurement process.

Sourcing and curating curriculum-aligned educational content.

On-premise data security and privacy

Key Resources

Core Technology: Offline AI models (e.g., Gemma, Mistral) and software platform

Human Capital: A team with deep expertise in both education/curriculum and the B2G sales and training process.

Technology: Proprietary, on-device AI models that ensure student data privacy and security (on-premise)

Value Propositions

For Principals (The Buyer): We provide an intuitive, easy-to-implement platform with a comprehensive training package, which

For Teachers (The User/Saboteur): We provide an AI-powered "Teacher's Toolbox" that reduces weekly lesson prep time by 3-5 h

FOR IT Dept. Fewer help desk tickets, easier maintenance, and no security headaches.

For Homeschooling Parents: We provide a complete, curriculum-aligned AI learning assistant that works offline, reducing scre

~~For DOC security officials, Axam provides a completely offline, auditable, and secure digital environment that is easier to m~~

Customer Relationships

university research labs and non-profits: partnerships: high-touch and community-based

High-Touch Onboarding: A dedicated process for principals and teachers, including on-site or virtual professional development

Automated & Community Support: In-platform guides for teachers and a support system for ongoing use.

Channels

Primary: Direct outreach to principals and school leaders via a targeted sales effort.

Critical Step: Becoming an approved, contracted vendor with the NYC Dept of Ed to navigate the ERMA procurement Platform

Homeschooling Networks/Communities

Customer Segments

The Principal & School Leadership Team (SLT) - Decision Maker:

Students (grades 6-12) - User

Saboteur: Tech-Resistant Teachers (passive resistance) and the IT Department (security veto).

Department Chairs, Superintendents, PTA leaders. - Influencers

Economic Buyer- The Principal (e.g. Title I funds) -

Cost Structure

Cost fo accessing school curricula to trian on

Sales and compliance cycle required to get through the security and governance committees.

Secondary Costs: Content licensing, marketing & sales expenses, cloud computing for AI model training.

Revenue Streams

Primary: Annual per-school site license, priced based on student enrollment.

Secondary (to test): A "freemium" model where a basic version is free, with a paid subscription for an advanced analytics da

~~Principals (autonomy over their school's budget, especially with Title I funding)~~

Agile Development Roadmap - Now, Next, Later

Start Small, Grow Big

"Do one thing exceptionally well, then expand"

MVP: One Subject (e.g., Algebra)

Students:
- Tutor/Explainer
- ELI5 Concepts

Teachers:
- Lesson Plans
- Differentiators
- Handout Creator

Key Goals:
- ✓ 30 min training max
- ✓ Security-ready
- ✓ Seamless adoption
- ✓ Focus on simplicity

System Integration

- Google Classroom
- Admin Dashboard
- More Subjects

Scale Goals:
- 10+ pilot schools
- DOE approval

Advanced Features

- Principal Analytics
- Workforce Score (AI)
- Specialized Markets
  - Justice Education
  - Special Needs
  - Corporate Partners

Global Expansion

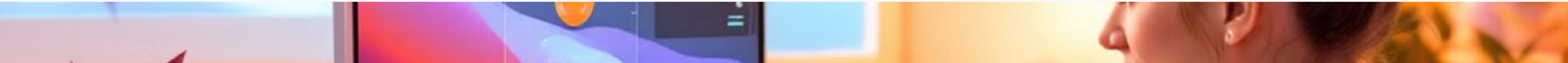
- International NGOs
- Refugee Education
- Advanced AI Suite

NOW
Months 1-3

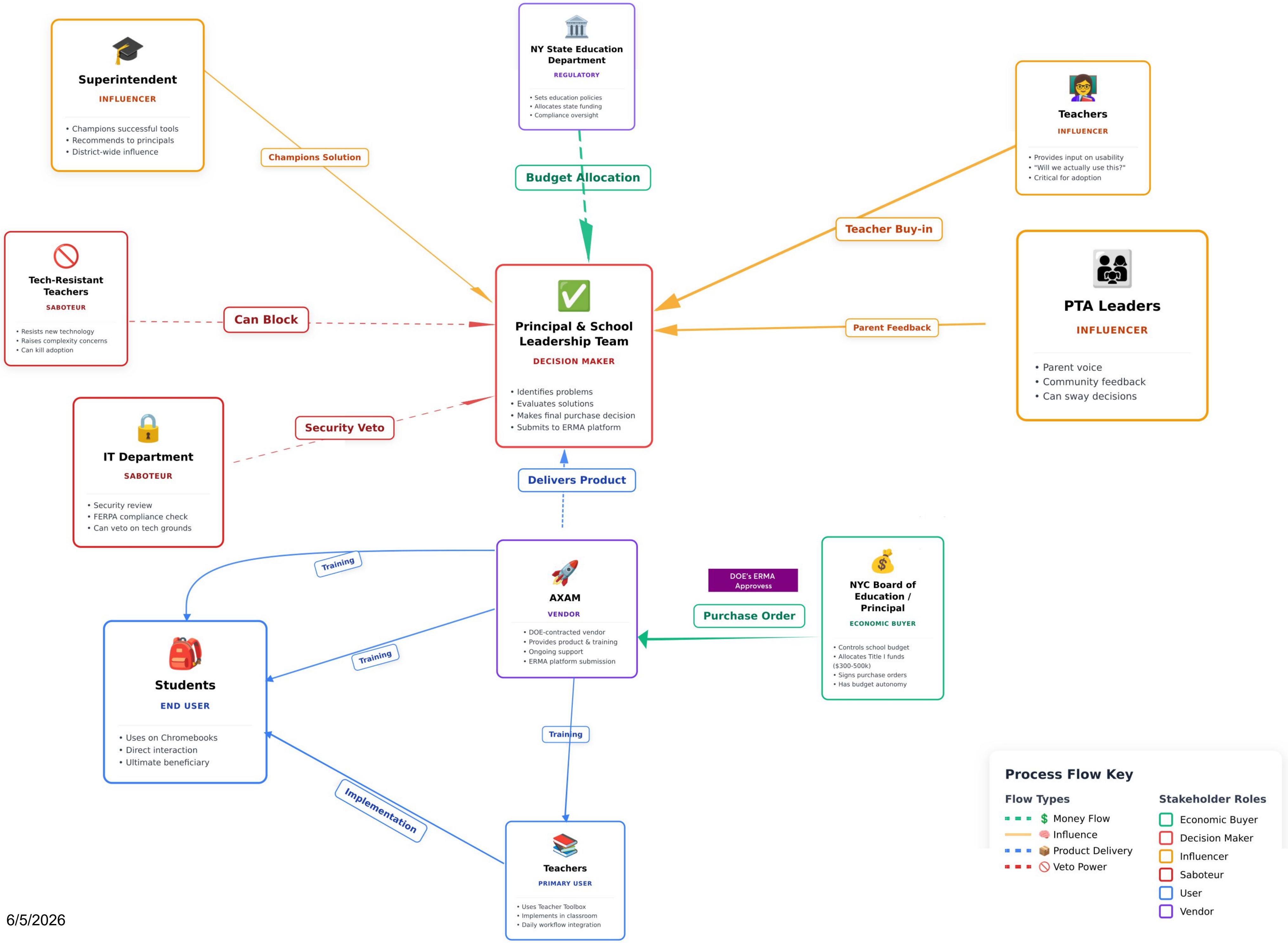
NEXT
Months 4-9

LATER
Year 2

VISION
Year 3+



The Axam Customer Ecosystem: Process Flow Diagram



The NYC Public School Ecosystem:

Users: Students, Teachers

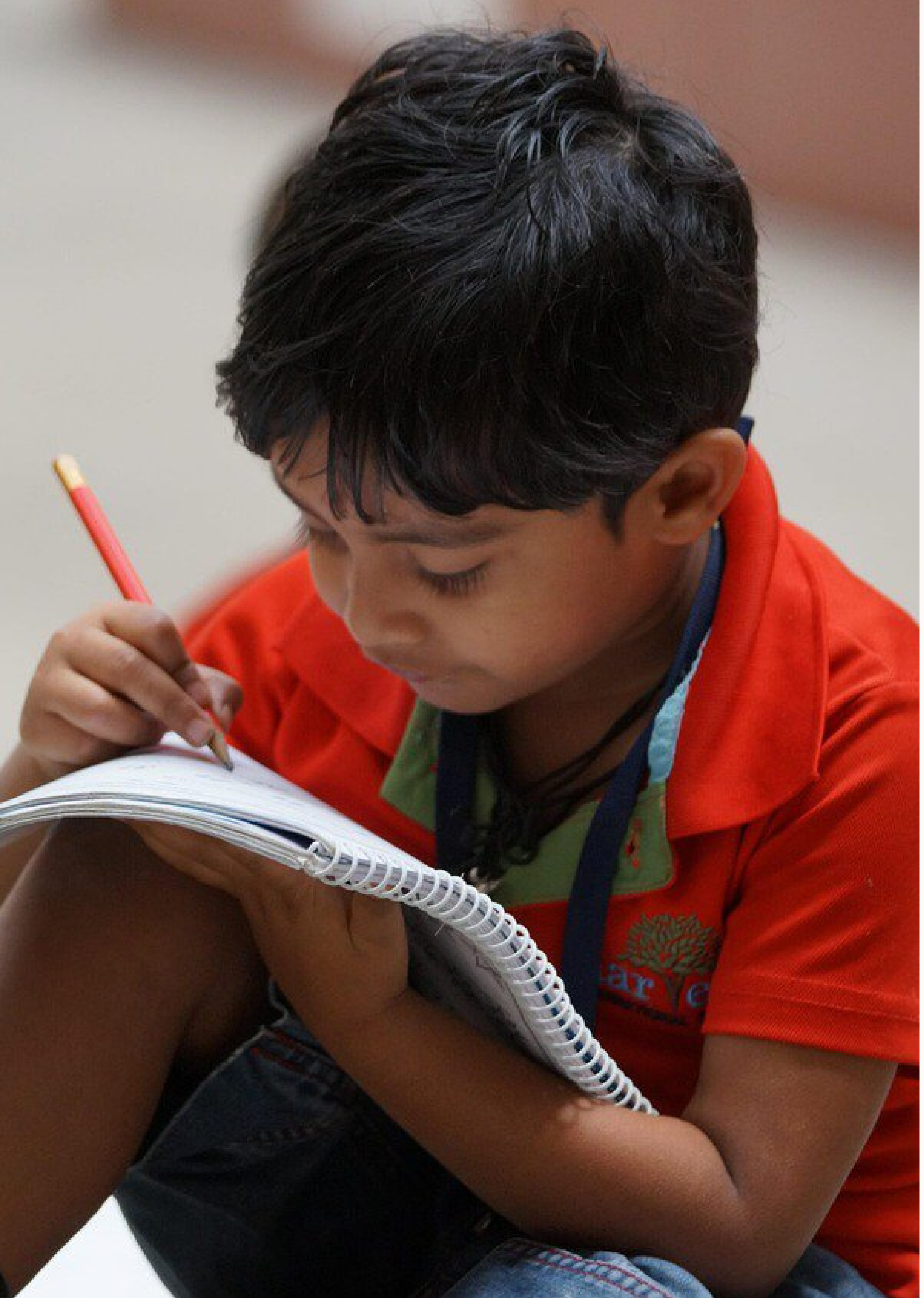
Economic Buyers: Principal (school budget), NYC Board of Education

Decision Makers: Principal, School Leadership Team

Influencers: Department Chairs, Superintendent, PTA Leaders

Saboteurs: Tech-resistant teachers, IT Security, Dept. of Corrections (for justice education)

Key Insight: Success requires navigating a complex web of stakeholders - ***it's a campaign***, not a simple sale.



IMPACT

The Impact We Envision:

- Students & Teachers having AI tutors to supplement their instruction
- Students in remote or under served areas accessing the same quality of knowledge as those in major cities
- Educational institutions reducing costs while improving learning outcomes.

TESTIMONIALS

"We trusted that the two of them would succeed in forging personal connections and drawing out equally engaging stories from a range of podcast guests.." - MIT OCW Team

“The problem a serious one, and the noble cause. We need a solution like this” – Monalissa, Human rights lawyer from Southern Africa

“Start small, its possible, first build your users and it will work, and when you have about 2000 users, approach Open AI” – Dr Mehdi Hassan-Proffessor Comp science and AI – Katz School

Next Steps

1. Secure Funding

Target impact investors and educational grants

- Apply for educational technology grants
- Pitch to impact-focused venture capital
- Explore partnerships with development organizations

2. Recruit Core Development Team

Build specialized AI/education team

- Hire AI/ML engineers with educational focus
- Onboard content validation experts

3. Finalize MVP

Focus on 3 core subjects module excellence

- Complete AI model training different Subjects
- Implement cross-validation system
- Speed up queries
- Add More multilingual capability
- Adding Audio/Speech
- Conduct user testing and refinement